

Anthony GX-Chen

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Education

- New York University** — Ph.D. Data Science (GPA 4.0/4.0) 2021 – ongoing
Advisors: Prof. Rob Fergus, Prof. Rajesh Ranganath
- McGill University / Mila** — M.Sc. Computer Science (Thesis) (GPA 4.0/4.0) 2019 – 2021
Advisors: Prof. Joelle Pineau, Prof. Blake Richards
Thesis: On Successor Representations for Value Learning: efficient credit assignment through implicit models. [lpdf](#)
- McGill University** — B.Sc. Major in Neuroscience (GPA 3.98/4.0) 2015 – 2019

Academic Experience

- New York University** — Doctoral Research Assistant 2021–present
- Advanced theoretical understanding of effect of RL on foundational generative models (LLMs, chemical language models). Used theory to design principled & simple methods for improving generation diversity and quality.
 - Investigated language model (LM) agents' inability to causally explore; developed [scalable test-time sampling](#) that doubles information gain and improves causal relationship identification accuracy by 55%.
 - Research in [model-based reinforcement learning \(RL\)](#) with semantic abstractions; developed algorithms for exploration and [efficient world model learning](#), enabling [>2x longer planning depth](#) than leading methods.
- McGill University / Mila** — Masters Research Assistant 2019–2021
- Research on [sample-efficient RL](#); combined future-predictive representations with λ -return, yielding [~20% lower value-learning error](#) and [~50% higher policy rewards](#) vs. TD.
 - Discovered connections between [successor representations](#) and [temporal credit assignment](#); presented as an oral at the Champalimaud Research Symposium (2021).
- Douglas Hospital, Montreal** — Research Assistant (Computational Neuroscience) 2017–2019
Supervised by Prof. Mallar Chakravarty
- Unsupervised categorization of behaviorally relevant brain sub-regions using multi-modal neuroimaging.
- University College London** — Research Assistant (Theoretical Neuroscience) Spring 2018
Supervised by Prof. Karl Friston
- Cognitive modeling of human behavior; developed variational Bayesian model of habit formation (MATLAB).

Industry Experience

- Google DeepMind** — Student Researcher Fall 2025
- [LLM RL post-training](#). Hosted by Ankit Anand and Gheorghe Comanici.
- Meta** — PhD SWE Intern Summer 2025
- Multimodal [LLM RL post-training](#) for video content understanding on the Brand Safety team.
 - Improved F1 score by [55%](#) over base models; [8%](#) better than larger prompt-engineered models.
- Microsoft** — Data Science Intern Summer 2019
- Industry research on information extraction using word embeddings for the Satori NLP team.
 - Improved unsupervised mention-extraction F1 by [>62% \(relative\)](#), focusing on biomedical domains.
- Microsoft** — Software Engineering Intern Summer 2018
- Built distributed cloud metrics pipeline (C#) for analytics; increased actionable insights by [>500%](#).
 - Enabled rapid analysis of millions of records for Microsoft Azure's data needs.

Selected Publications

1. **GX-Chen, Anthony**, Jatin Prakash, Jeff Guo, Rob Fergus, Rajesh Ranganath. "KL-Regularized Reinforcement Learning is Designed to Mode Collapse" *Under Submission* (2025). [\[arXiv\]](#)
2. **GX-Chen, Anthony**, Jatin Prakash, Rob Fergus, Rajesh Ranganath. "Reverse-KL Reinforcement Learning Can Sample From Multiple Diverse Modes" *NeurIPS Workshop on Foundations of Reasoning in Language Models* (2025).
3. **GX-Chen, Anthony**, Dongyan Lin*, Mandana Samiei*, D. Precup, B. Richards, R. Fergus, Kenneth Marino. "Language Agents Mirror Human Causal Reasoning Biases. How Can We Help Them Think Like Scientists?" *COLM* (2025). [\[pdf\]](#)
4. **GX-Chen, Anthony**, Kenneth Marino, Rob Fergus. "Efficient Exploration and Discriminative World Model Learning with an Object-Centric Abstraction." *ICLR 2025*. [\[arXiv\]](#)
5. **GX-Chen, Anthony***, Dongyan Lin*, Mandana Samiei*. "Testing causal hypotheses through Hierarchical Reinforcement Learning." *NeurIPS Workshop on Intrinsically Motivated Open-ended Learning* (2024). [\[OpenReview\]](#)
6. Zhang, Wancong, **Anthony GX-Chen**, Vlad Sobal, Yann LeCun, Nicolas Carion. "Light-weight probing of unsupervised representations for Reinforcement Learning." *Reinforcement Learning Conference*, Oral (2024). [\[arXiv\]](#)
7. **GX-Chen, Anthony**, Veronica Chelu, Blake A. Richards, Joelle Pineau. "A Generalized Bootstrap Target for Value-Learning, Efficiently Combining Value and Feature Predictions." *AAAI* (2022). [\[arXiv\]](#)
8. **GX-Chen, Anthony**, Veronica Chelu, Blake Richards, Joelle Pineau. "Lambda Successor Return Error." *NeurIPS Workshop on Biological and Artificial Reinforcement Learning* (2020). [\[pdf\]](#)

Other Publications

1. **GX-Chen, Anthony**, David Benrimoh, Thomas Parr, and Karl J. Friston. "A Bayesian Account of Generalist and Specialist Formation Under the Active Inference Framework." *Frontiers in Artificial Intelligence* 3 (2020).
2. Patel, Raihaan, Christopher J. Steele, **GX-Chen, Anthony**, Sejal Patel, ..., and M. Mallar Chakravarty. "Investigating microstructural variation in the human hippocampus using non-negative matrix factorization." *NeuroImage* 207 (2020): 116348.
3. Trakadis, Yannis J., Sameer Sardaar, **GX-Chen, Anthony**, Vanessa Fulginiti, and Ankur Krishnan. "Machine learning in schizophrenia genomics, a case-control study using 5,090 exomes." *American Journal of Medical Genetics Part B* (2019).

Awards

NSERC (Canada) — Canada Graduate Scholarship, Doctoral (CGS-D) (\$105,000) Offered to retain high-calibre doctoral students at Canadian institutions.	Sept 2021
NSERC (Canada) — Postgraduate Scholarship, Doctoral (PGS-D) (\$63,000) Selected on the basis of research ability, communication, interpersonal, and leadership abilities.	April 2021
Fonds de Recherche du Québec — FRQNT Master's Scholarship (\$17,500)	April 2020
Centre UNIQUE & McGill University — Master's Excellence Scholarship (\$10,000)	Feb. 2020
Canadian Institutes of Health Research — Canada Graduate Scholarship, Master's (\$17,500)	Aug. 2019

Teaching & Mentoring

Courses Taught / TA

DSGA 3001: Learning from Small Data (NYU Center for Data Science) — Section Leader (TA)	Fall 2024
Intro to Reinforcement Learning (NYU Pathways to AI Program) — Instructor Taught a two-day course to undergraduates on MDP theory and policy-gradient algorithms.	July 2024

Students Advised

Skylar Wang — Yale Undergraduate Student Advised senior thesis research project	Fall 2025
Ronak Agarwal — NYU Summer Undergraduate Student Research Project	Fall 2024
Carla Garcia Medina — NYU Undergraduate Student Advised senior research project; now Research Engineer at Google LLC.	2022–2023

Skills

- **General Scientific Computing:** Bash, Numpy, SciPy, Pandas, Jupyter, Git, Dask, SQL, MATLAB, Java, C#.
- **Machine Learning:** PyTorch, JAX, Scikit-learn, TensorFlow.
- **Distributed High-Throughput (Reinforcement) Learning:** Torch DDP, Slurm, Sample-factory, multiprocessing.
- **Foundational Models, Fine-tuning and Inference:** HuggingFace, vLLM, verl, OpenAI API, mixed-precision training.